

$$c_{\text{liquid}} = 2c_{\text{solid}}$$

10. Ans: B

$$C_{rms} = \sqrt{\frac{3RT}{M}} \propto \sqrt{T} \propto \sqrt{PV} \quad \text{since } PV = nRT$$

$$\frac{C_{rms2}}{C_{rms1}} = \sqrt{\frac{P_2 V_2}{P_1 V_1}} = \sqrt{\frac{2(2)}{1(1)}} = 2$$

11. Ans: A